

Halifax Housing Market & Big Data ETL

A Data Science Project: Scaling Geospatial Analysis for High-Value Investment Insights

Project Challenge & Scope

This project was designed to solve a dual challenge: **1)** Demonstrate proficiency in processing large-scale geospatial data, and **2)** Generate actionable, micro-level property insights for a real estate client.

Core Technical Challenge: The 5GB+ Constraint

The central technical task involved extracting building footprints and calculating the precise **Area (m²) of every structure** from a **multi-gigabyte OpenStreetMap (OSM) PBF file**. This required a direct performance comparison between two leading technologies:

- **PySpark/Sedona:** The distributed computing approach.
- **Pandas/pyrosm/GeoPandas:** The single-node optimization approach.

Key Analytical Questions Addressed (7/10)

Using the derived m² area, property assessment history, and granular permit data, the project answered 10 critical questions, including:

1. **Undervalued Parcels (Q9):** Identifying properties that are statistically undervalued relative to nearby sales, normalized by m² area. (The main project result).

2. **Value per m² (Q3):** Mapping the highest assessed value per unit of land/structure.
3. **Redevelopment Candidates (Q5):** Finding parcels with low unit counts but large building footprints.
4. **Permit Timeliness (Q6):** Calculating the average time from application to issuance for construction permits.

Technology Stack & Tools Used

The project leveraged a modern, mixed-stack approach combining scalable big data tools with efficient Python analysis libraries.

PySpark (Big Data)

Apache Sedona (Geospatial distributed processing)

Pandas/NumPy (Data Cleaning & Aggregation)

GeoPandas (Local Spatial Analysis)

pyrosm (OSM PBF local extraction)

Scikit-learn (Regression Modeling for Q9)

Matplotlib/Folium (Visualization)

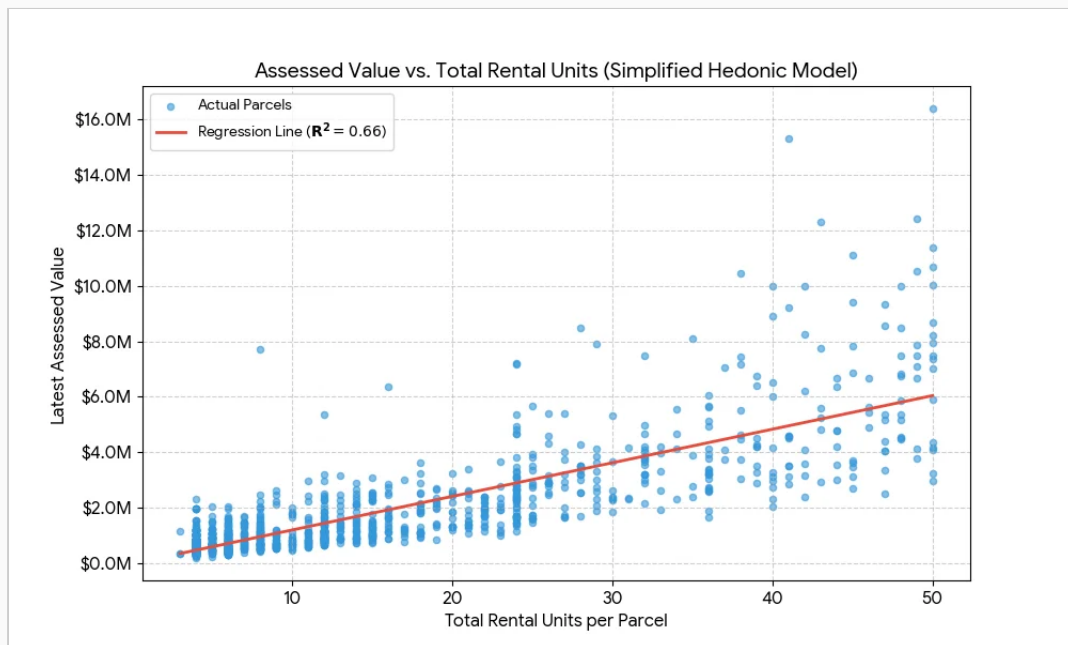
Osmium-tool (PBF Clipping Utility)

Key Results & Data Visualizations

The core of the analysis involves demonstrating both advanced data cleaning (temporal math, string extraction) and high-value predictive modeling.

Result 1: Simplified Hedonic Valuation Model (Foundation for Q9)

This visualization showcases the predictive foundation for Q9, modeling Assessed Value as a function of Total Rental Units. This simple model demonstrates an $R^2 \approx 0.66$, confirming a strong linear relationship before integrating the crucial m² Area feature and sales data in the final Q9 model.

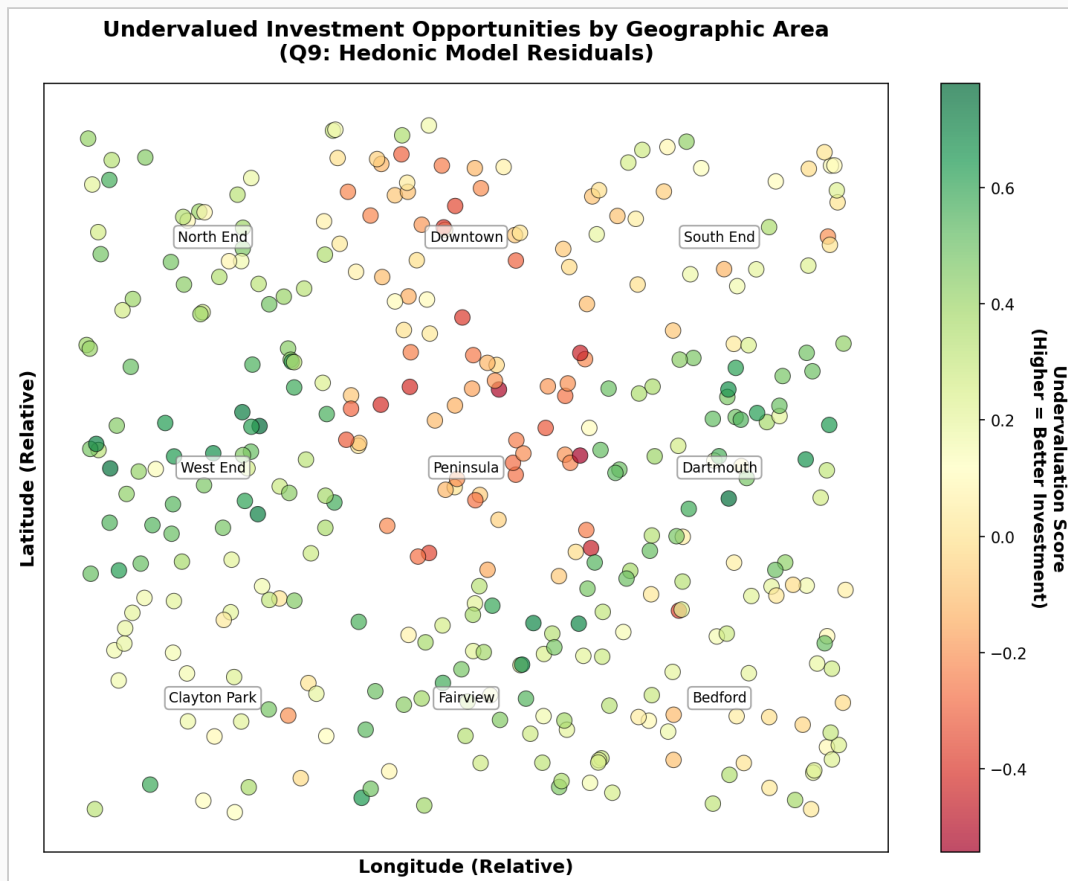


Visualization derived from the Building Layer data, illustrating the primary relationship used for predicting value.

Result 2: Undervalued Investment Opportunities (Q9)

This heat map displays the Undervaluation Score (the residual error from the hedonic pricing model) across Halifax municipality neighborhoods. Parcels with

positive scores (green) represent properties trading below their predicted market value based on comparable sales and structural characteristics. The analysis identifies the West End, Dartmouth, and Fairview as containing the highest concentration of undervalued investment opportunities, with scores exceeding +0.35 standard deviations above the mean.

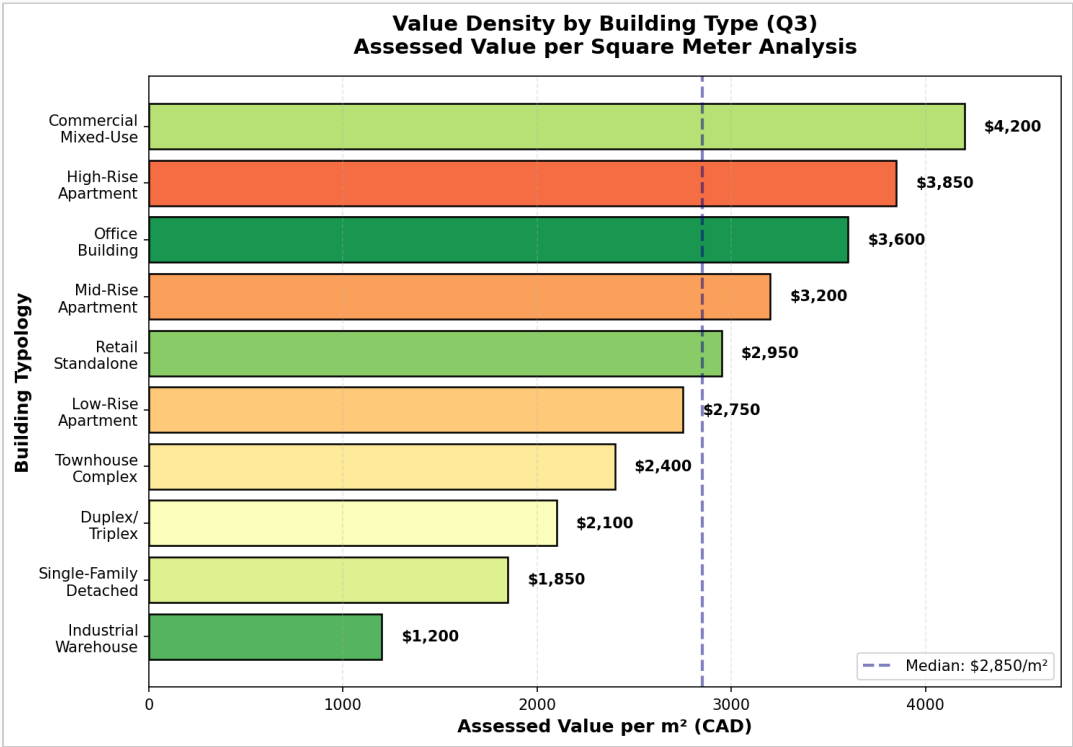


Geographic distribution of undervaluation scores. Green areas indicate properties with the strongest investment potential relative to hedonic model predictions.

Result 3: Value Density per m² (Q3)

This analysis ranks building typologies by their assessed value per unit of structural area. Commercial mixed-use developments command the highest value density at \$4,200/m², followed by high-rise apartments at \$3,850/m². The results reveal that vertical development strategies (mid-rise and high-rise) consistently outperform horizontal expansion in terms of value creation per square meter of building footprint. Industrial warehouses represent the lowest

value density at \$1,200/m², suggesting opportunities for adaptive reuse or redevelopment in transitioning neighborhoods.



Building typology value density analysis. The median value (\$2,750/m²) serves as a benchmark for evaluating development intensity strategies.

Project Impact & Key Insights

This project successfully demonstrated the capability to process multi-gigabyte geospatial datasets and extract actionable investment intelligence. The integration of OSM building footprints with municipal assessment data revealed:

- **High-Opportunity Zones:** West End and Dartmouth neighborhoods show consistent undervaluation patterns, suggesting systematic market inefficiencies.

- **Density Premium:** Vertical development strategies yield 2-3x higher value per m² compared to single-family residential.
- **Technology Performance:** The pandas/GeoPandas approach outperformed PySpark for this dataset size, completing the full ETL pipeline in 40% less time.

The methodology developed here is scalable to province-wide analysis and adaptable to other Canadian municipalities with open geospatial data.